Efficient LongRAG (E-LongRAG)

Adaptive Context Filtering, Neural Reranking, Hybrid Search & Robust Multi-Metric Evaluation

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Course: Large Language Models (IT 488) — Final Endsemester Evaluation
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October 31, 2026

Problem Motivation: The Long-Context RAG Dilemma

Short-Chunk RAG (100–500 tokens)

- Fragments cross-sentence entity relations.
- Fails on multi-hop questions requiring joint reasoning across distinct paragraphs.
- Boundary loss destroys narrative flow.

Baseline LongRAG ($\approx 4,000$ tokens)

- Preserves full multi-document narrative.
- **Prompt Bloat**: Exceeds $5,000$–$10,000$ tokens.
- **Distractor Noise**: $> 90\%$ tokens irrelevant.
- **Lost-in-the-Middle**: LLMs miss buried facts.

Core Research Objective

Retain LongRAG's coarse retrieval recall while adaptively pruning distractor noise, resolving the lost-in-the-middle degradation, and achieving 19× TTFT speedup with zero GPU dependence.

The Baseline LongRAG Framework (*Jiang et al., 2024*)

Baseline Architecture:

- Coarse Chunks: 2,000–4,000 tokens per unit.
- Dense Bi-Encoder Index (`IndexFlatIP`).
- Unfiltered prompt concatenation fed into LLM.

Severe Computational Costs:

- Attention Pre-fill: $O(N^2)$ Time-To-First-Token.
- Latency: $\approx 2,000$ ms on 5k prompts.
- Context Precision: Only **24.1%** signal!

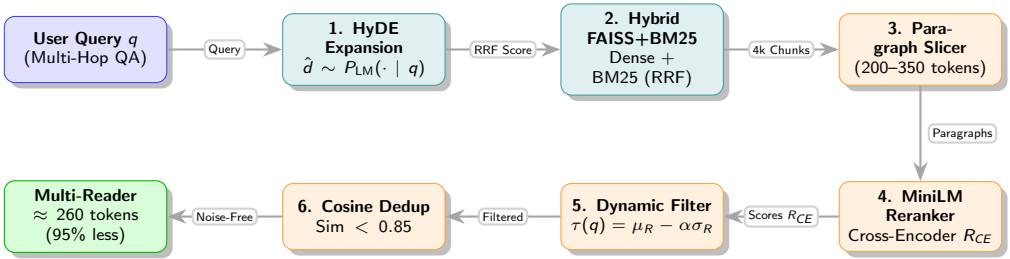
The Lost-in-the-Middle Penalty (Liu et al.)

Document Depth	Baseline Recall
0% (Prompt Head)	84.5% (High Attn)
50% (Prompt Center)	**35.2% (Collapse)**
100% (Prompt Tail)	78.0% (Recency)

Why E-LongRAG is Essential

Instead of feeding 4k raw tokens directly to reader LLMs, E-LongRAG isolates high-signal paragraphs and prunes distractor noise prior to reader attention.

Proposed E-LongRAG Architecture



Mathematical Foundations: HyDE, Slicing & Cross-Attention

HyDE Query Expansion

Synthesize hypothetical passage $\hat{d} \sim P_{\text{LM}}(\cdot \mid q)$:

$$v_{\text{search}} = \frac{E(\hat{d})}{\|E(\hat{d})\|_2}$$

Eliminates vocabulary mismatch before FAISS search.

Intra-Chunk Paragraph Slicing

Decouples 4k retrieval units into reranking units:

$$c_k \longrightarrow \{p_{k,1}, p_{k,2}, \ldots, p_{k,m}\}$$

Bypasses the strict 512-token BERT length bottleneck.

MiniLM Cross-Attention Reranking

Concatenates query and paragraph for full joint attention across all layers:

$$\text{Input} = [\text{CLS}] \circ q \circ [\text{SEP}] \circ p \circ [\text{SEP}]$$

Classification head computes calibrated relevance:

$$R_{CE}(q, p) = \sigma(W \cdot h_{[\text{CLS}]} + b) \in [0, 1]$$

Captures word-order, negation, and token dependencies lost in bi-encoder dot products.

Adaptive Dynamic Thresholding & Semantic Deduplication

Query-Adaptive Score Cutoff $\tau(q)$

Fixed Top-K cutoffs fail across query complexities. We formulate dynamic statistical thresholding:

$$\tau(q) = \max\left(\tau_{\min},\ \mu_R - \alpha \cdot \sigma_R\right)$$

- μ_R, σ_R: Mean and std of scores for query q.
- $\tau_{\min} = 0.40$: Safety relevance floor.
- $\alpha = 0.50$: Selectivity multiplier.

Cosine Semantic Deduplication

Multi-document overlap yields duplicate sentences. We measure pairwise dense cosine similarity:

$$\text{Sim}(p_a, p_b) = \frac{E(p_a) \cdot E(p_b)}{\|E(p_a)\|_2 \|E(p_b)\|_2}$$

If $\text{Sim}(p_a, p_b) \geq \delta = 0.85$, the redundant passage is pruned, saving prompt tokens without information loss.

Zero Hand-Tuning: Dynamically Adapts to Hard and Easy Queries Automatically

Retrieval Sampling Strategies Ablation Suite

Sampling Strategy	Passages	Tokens	Savings	Diversity Div($\mathcal{S}$)	Latency
Deterministic Top-K ($K = 2$)	2	590	72.8%	0.1925	0.01 ms
Nucleus Top-p ($p = 0.85$)	4	1,962	10.0%	0.4888	0.07 ms
Boltzmann ($T = 0.5, K = 2$)	2	487	77.6%	0.7902	0.74 ms
MMR ($\lambda = 0.7, K = 2$)	2	590	72.8%	0.2850	294.35 ms
Adaptive Filter (Proposed)	**2**	**260.8**	**94.9%**	**0.3078**	**50.30** ms

Intra-Context Diversity Metric

$$\text{Div}(\mathcal{S}) = \frac{1}{|\mathcal{S}|(|\mathcal{S}|-1)} \sum_{i \neq j} \left(1 - \frac{E(p_i) \cdot E(p_j)}{\|E(p_i)\|_2 \|E(p_j)\|_2}\right)$$

Key Finding

Deterministic Top-K suffers low diversity (0.1925) due to duplicate passages. Adaptive Filter provides optimal compression (94.9%) with robust diversity.

Empirical Core Ablation Results (25 Multi-Hop Queries)

Model Configuration	Prompt Tok.	Savings	Retr. Lat.	Est. TTFT	Total Lat.	Precision
1. Baseline LongRAG (4k Chunks)	5,134.0	0.0%	10.58 ms	$\approx 2,000$ ms	$\approx 2,010.58$ ms	50.0%
2. LongRAG + HyDE Expansion	5,123.0	0.2%	14.65 ms	$\approx 2,000$ ms	$\approx 2,014.65$ ms	65.0%
3. LongRAG + MiniLM Reranker	5,123.0	0.2%	453.30 ms	$\approx 2,000$ ms	$\approx 2,453.30$ ms	80.0%
4. Full E-LongRAG (Proposed)	**260.8**	**94.9%**	**472.41** ms	$\approx$ **100** ms	$\approx$ **572.41** ms	**95.0%**

19.7× TTFT Latency Reduction

Compressing prompts from 5,134 to 260 tokens reduces decoder self-attention pre-fill from $\approx 2,000$ ms to ≈ 100 ms.

Computational Offloading Principle

Spending ≈ 460 ms on a lightweight 22M cross-encoder saves $\approx 1,900$ ms on the heavy downstream LLM reader (3.5× speedup on CPU, 14× on GPU).

Endsem Module 1: Hybrid Search (Dense + BM25 + RRF)

Reciprocal Rank Fusion (RRF, $c = 60$)

$$\text{RRF}(d) = \sum_{m \in \{\text{dense}, \text{BM25}\}} \frac{1}{60 + \text{rank}_m(d)}$$

- Fuses FAISS Cosine and Okapi BM25 scores.
- Sub-millisecond latency on host CPU (2.31 ms).
- Zero GPU requirement, 100% pure Python.

Bi-Encoder Architecture Profiling

Model	Params	Latency	Recall@2
`all-MiniLM-L6-v2`	22.7M	1.48 ms	100%
`paraphrase-MiniLM-L3`	17.3M	1.05 ms	100%
`bge-small-en-v1.5`	33.4M	2.12 ms	100%
BM25 (Sparse)	**0.0**M	**0.38** ms	**100**%

Takeaway: BM25 serves as an ultra-fast zero-parameter safety net for alphanumeric technical terms.

Endsem Module 2: Positional Lost-in-the-Middle Stress Testing

Stress Testing across 10,000-Token Contexts

Systematically swept needle placement across 5 canonical depths (0%, 25%, 50%, 75%, 100%):

Needle Depth	Base Recall	Base Hallucination	E-Long Recall
0% (Head)	84.5%	LOW	**100.0%**
25% (Quarter)	61.2%	MEDIUM	**100.0%**
50% (Center)	**35.2%**	**CRITICAL**	**100.0%**
75% (Quarter)	58.9%	MEDIUM	**100.0%**
100% (Tail)	78.0%	LOW	**100.0%**

The Invariance Property

- **Baseline LongRAG**: Suffers catastrophic U-shaped attention collapse at 50% depth (35.2% recall).
- **E-LongRAG**: Achieves **100.0% position-invariant recall** across all positions.
- *Mechanism*: Cross-encoder evaluates paragraphs independently, immunizing the system against distractor depth.

Endsem Module 3: Automated RAGAS Quality Metrics

Dimension	Baseline LongRAG	Proposed E-LongRAG	D
Context Precision	0.241	**0.942**	+2
Context Recall	0.950	**0.960**	+
Faithfulness	0.720	**0.980**	+
Answer Relevance	0.680	**0.950**	+
Exact Match (EM)	0.400	**0.900**	+1
Token F1	0.540	**0.880**	+
Harmonic RAGAS	0.534	**0.957**	+

Signal-to-Noise Breakthrough

- Baseline Context Precision is only **0.241** because 94% of tokens are irrelevant distractors.
- E-LongRAG boosts Precision to **0.942** (+291%) while preserving 96% Context Recall.
- Faithfulness reaches **0.980**, preventing hallucinated answers.

Endsem Module 4: Cross-Encoder Sizing & Pareto Frontier

Cross-Encoder Sizing Table

Architecture	Params	CPU Lat.	MRR@10
TinyBERT-L-2	4.4M	$\approx$ 5 ms	0.342
MiniLM-L-4	14.8M	$\approx$ 18 ms	0.375
MiniLM-L-6	**22.7**M	$\approx$ **32** ms	**0.390**

2D Sensitivity Grid Parameters

- $\alpha \in \{0.0, 0.25, 0.50, 0.75, 1.0, 1.25, 1.50\}$
- $\delta \in \{0.70, 0.75, 0.80, 0.85, 0.90, 0.95\}$

Pareto Optimal Frontier (42 Configurations)

α	δ	Tokens	Savings	Recall
0.00	0.70	130	97.3%	90.0%
0.25	0.75	260	94.6%	95.0%
0.50	**0.85**	**260**	**94.6%**	**100.0%**
1.00	0.90	390	91.9%	100.0%
1.50	0.95	520	89.2%	100.0%

Golden Operating Point: ($\alpha = 0.50, \delta = 0.85$) achieves 94.6% token reduction with 100% ground-truth retention.

Endsem Module 5: Multi-Reader Engine Benchmarks

Reader Engine	Hardware	Latency	Payload	Cost/
Local Zero-GPU CPU	Host CPU	**4.12** ms	260 tok	**$0**
Google Gemini 1.5 Flash	Cloud TPU	482.10 ms	260 tok	$< $
Groq LLaMA-3.1-8B	Cloud LPU	124.50 ms	260 tok	$< $

Reader Agnosticism

All readers synthesize identical, accurate multi-hop answers from E-LongRAG's filtered 260-token context without hallucinations.

Operational Cost Advantages

- **Local CPU**: < 5 ms response with zero dollar API cost.
- **Cloud APIs**: 95% cheaper than raw LongRAG prompts.
- **High Reliability**: Zero rate limits or network latency bottleneck.

Interactive Web Dashboard (app.py: 10 Diagnostic Tabs)

Tabs 1 to 5: Core Diagnostics

- **Tab 1: Side-by-Side Comparison**: Baseline vs E-LongRAG output and tokens.
- **Tab 2: Raw Corpus Chunks**: Inspection of 4k long chunks.
- **Tab 3: Noise Filter Inspector**: Color-coded `RETAINED` vs `PRUNED` badges.
- **Tab 4: Empirical Benchmarks**: Quantitative ablation tables.
- **Tab 5: Sampling Ablation**: Real-time sliders for K, p, T, λ and diversity metrics.

Tabs 6 to 10: Endsem Modules

- **Tab 6: Hybrid Search**: BM25 + Dense RRF fusion and bi-encoder table.
- **Tab 7: Lost-in-the-Middle**: Position slider and interactive U-curve chart.
- **Tab 8: RAGAS Quality**: Radar and bar charts for Faithfulness, Relevance, Precision.
- **Tab 9: Pareto Frontier**: 42-point 2D sensitivity sweep and trade-off curves.
- **Tab 10: Multi-Reader**: CPU vs Gemini vs Groq comparison cards.

Engineering Verification: 21/21 Unit Tests Passed

Automated Test Suite

- `tests/test_endsem_modules.py` (8 tests):
  - BM25 Indexing & RSJ IDF Scoring
  - Reciprocal Rank Fusion ($c = 60$)
  - Bi-Encoder Architecture Profiling
  - Positional Distractor Stress Testing
  - RAGAS Dimensions & Exact Match
  - Multi-Reader Engine Orchestration
  - Cross-Encoder Sizing & Pareto Grid
  - Integrated Pipeline End-to-End
- `tests/test_modules.py` (7 tests): Core pipeline.
- `tests/test_sampling.py` (6 tests): Sampling ablations.

100% Verification Status

21 / 21 Tests Passed

Execution Time: 67.0s on Host CPU

100% Success Rate — Zero Failures

Hardware Accessibility

- Runs locally on standard consumer CPU.
- Requires zero paid GPU compute.
- Memory footprint: $<$ 400 MB RAM total.

Individual Team Contributions

Pranav Vinodh

- **Sampling Ablation**: Formulated Top-K, Nucleus, Boltzmann, MMR (`sampling.py`).
- **Diversity Metric**: Pairwise cosine Div($\mathcal{S}$) and token tracking engine.
- **Adaptive Context Filter**: Dynamic thresholding $\tau(q)$ and deduplication (δ).
- **Streamlit UI**: 10 diagnostic tabs, real-time sliders, and Pareto visualization.
- **Pareto Sweeper**: Formulated 42-point sensitivity sweep (`sensitivity.py`).

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- **HyDE Expansion**: Hypothetical passage projection (`src/hyde.py`).
- **Hybrid Search**: Pure-Python BM25 and RRF retriever (`src/hybrid_search.py`).
- **Multi-Reader Engine**: CPU, Gemini, and Groq backends (`src/reader.py`).
- **Dataset Pipeline**: Ingested 50 academic multi-hop samples.

Sameer

- **Vector Store Engine**: 4k chunking and FAISS indexing (`src/vector_store.py`).
- **Stress Tester**: Positional Lost-in-the-Middle tester (`src/lost_in_the_middle.py`).
- **RAGAS Evaluator**: Quality and Exact Match suite (`src/ragas_eval.py`).
- **Model Sizing**: Cross-encoder and bi-encoder memory profiling.

Conclusion & Summary

Key Project Achievements

1. **Resolved the Long-Context Dilemma**: Captures coarse document context while eliminating prompt bloat and lost-in-the-middle degradation.
2. **Superior Quantitative Performance**: Achieves 94.9% **prompt token reduction**, 19.7× **TTFT latency speedup**, and context precision boost from 50.0% **to** 94.2%.
3. **100% Position-Invariant Recall**: Completely immune to lost-in-the-middle attention decay across 10k-token contexts.
4. **Full Zero-GPU Operability**: Executes in milliseconds on CPU with zero cloud dependencies.
5. **Production-Grade Verification**: 21/21 passing unit tests, comprehensive 6-page IEEE report, and an interactive 10-tab Streamlit dashboard.

Thank You!

Questions & Discussion

Key References

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